Optimizing Retrieval-Augmented Generation for Construction Management: An Ablation Study on Embedding and Generative Models

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Abstract

This research presents a comprehensive ablation study of Retrieval-Augmented Generation (RAG) systems for question-answering in the construction management domain. Utilizing the Wendel specifications document as the foundational knowledge base, we curated a ground truth dataset consisting of approximately 300 domain-specific question-answer pairs that cover a wide spectrum of construction processes, codes, and regulations. Our approach systematically varied three components: embedding models, generative models, and generation temperature, to assess their effects on the accuracy and quality of generated responses. For retrieval, we compared the IBM Granite Embedding Model (278M) with nomic-embed-text (137M), while for generation, we evaluated three models (Gemma3 1b, Gemma3 4b, and DeepSeek-R1-Distill-Qwen-7B) across temperature settings of 0.1, 0.5, and 0.8. Quantitative evaluation using BLEU and ROUGE metrics revealed that the IBM Granite Embedding Model consistently outperformed its counterpart, and that an optimal balance between precision and recall was achieved at a moderate temperature setting of 0.5, particularly when paired with the Gemma3 1b model (BLEU: 0.1252, ROUGE: 0.3178). Our findings support claims in recent literature that carefully tuned RAG configurations can help mitigate hallucinations and improve information retrieval accuracy in highly specialized domains.
Github Repository Link: Group 6 NLP Project
Data Link: Group 6 NLP Project Data

1. Introduction

Construction engineering, safety, and management form the backbone of modern infrastructure development and maintenance. With increasing complexity in construction projects and stringent regulations designed to ensure worker safety and structural reliability, traditional methods of construction project management and monitoring are often strained. Traditional methods have long relied on manual documentation, periodic inspections, and extensive human oversight. However, as projects grow in scale and complexity, with increased integration of advanced technologies, simultaneous multi-disciplinary operations, and tighter safety protocols, these conventional approaches struggle to keep pace. Additionally, stringent regulations mandate continuous compliance checks and real-time reporting to ensure both worker safety and structural reliability, placing considerable strain on methods that depend on sporadic, human-driven processes.

Recent advances in artificial intelligence (AI), particularly in generative AI, offer promising solutions to these challenges by automating routine tasks, improving decision-making processes, and providing insights from vast amounts of unstructured data. In this context, the integration of generative AI into construction management processes is emerging as a critical tool to improve safety compliance, streamline operations, and ultimately reduce costs and risks.

Retrieval-augmented generation (RAG) models have gained significant traction across various domains due to their ability to combine the strengths of information retrieval with the advanced generation capabilities of large language models (LLMs). These models effectively search for relevant information within large databases and utilize it to generate coherent and contextually accurate responses. Their utility spans numerous industries, from healthcare and finance to legal and educational sectors, by creating systems that can answer complex queries, summarize extensive documents, and support real-time decision-making.

The potential of RAG models also extends to the construction domain. Given the vast array of specifications, regulations, and technical documentation prevalent in construction engineering, integrating RAG into construction safety and management systems can revolutionize how projects are planned, executed, and monitored. By linking specific sections of technical documents to detailed Q&A pairs, these models can significantly improve the precision and relevance of the generated responses. This integration not only aids in compliance and quality assurance but also helps to translate complex engineering data into practical recommendations.

In our project, we have developed a comprehensive RAG system specifically designed for the construction safety and management domain. Our work involved creating a ground truth dataset from the Wendel specifications document, implementing retrieval methods using state-of-the-art embedding models, and generating responses with advanced generative models. A few of the major contributions of the project are as follows.

- The design and implementation of a RAG system tailored to construction safety and management, addressing domain-specific challenges.
- A detailed ablation study analyzing the impact of different embedding and generative model configurations on system performance.
- Evaluation using BLEU and ROUGE metrics to validate the quality and accuracy of the system's outputs.

The remainder of this report is structured as follows. Section 2 details the literature review, followed by Section 3, which describes the dataset utilized in the study. Section 4 focuses on the experimental methodology, while Section 5 discusses the results of the ablation study performed. Section 6 concludes the report with final remarks and future work recommendations.

2. Literature Review

Current research in construction management and civil engineering demonstrates progress in automating construction scheduling using large language models (LLMs) and reinforcement learning, notably CONSTRUCTA [1]. The framework utilizes predefined prompt categories and task-specific prompts for data collection, as well as specific tasks such as Automated Planning (AP), Missing Value Prediction (MVP), and Dependency Analysis (DA). The methodology of CONSTRUCTA revolves around 3 key components:

- Static Retrieval-Augmented Generation (SRAG or Static RAG): SRAG comprises local and global static RAG, wherein the local static RAG is responsible for providing precise definitions for construction specific terms, and the latter is responsible for broader domain knowledge from resources like textbooks and manuals using embeddings and chunk based retrieval.
- Contextualized Knowledge RAG (Knowledge RAG or KRAG): This component incorporates the expertise of architects by dynamically sampling context-sensitive information from a dependency graph of activities. It aggregates hierarchical context (nodes within the same Work Breakdown Structure up to two levels), first-order context (direct predecessors and successors), and sequential context (predecessors and successors up to three hops) for a target activity. The combined context is then used to retrieve relevant embeddings from the knowledge database for construction scheduling.
- Construction Reinforcement Learning from Human Feedback(RLHF): This component aligns the outputs of LLMs with expert feedback to improve their understanding and produce human-aligned scheduling decisions. It involves a Plan Agent that combines task-specific details with retrieved context and rules, and an expert agent that uses feedback and memorized domain knowledge to ensure that schedules align with project requirements.

Another initiative focuses on ensuring reliable retrieval of information to address the evolving technical standards in construction engineering and the need for engineers to access specialized knowledge bases. We can observe an innovative approach to enhance the performance of the traditional cosine similarity-based retrieval by employing Entropy-Optimized Dynamic Text Segmentation (EDTS) [2]. The EDTS algorithm involves counting word frequencies, splitting text into sentences, and then iteratively calculating conditional entropy as the context window expands to the left and right of each sentence to determine the optimal chunk boundaries. The proposed method employing EDTS aims to ensure information clarity and predictability within limited chunk lengths by dynamically partitioning text into semantically coherent chunks.

An initiative to develop a generative question-answering advisor for technical documents in the Architecture, Engineering, and Construction (AEC) domain aims at reducing the hallucinations in LLMs [3]. The architecture includes chunking of cleaned and tokenized data, followed by vectorization and storage in a vector database allowing fast cosine similarity based retrieval.

Special focus is placed on engineering high-quality prompts to the generative model. Incorporating design patterns like Persona Pattern and Format Template Pattern, and advanced techniques like Chain of Thought (CoT) and Few-shot Learning, these prompts guide the LLM to accurately interpret user queries. The developed advisor was evaluated by ten AEC experts who compared its performance against traditional search methods: Google search and the agency's Technical Documents Database (TDD). The evaluation metrics included Comprehensiveness, Accuracy, Relevance, Clarity, Conciseness, and Search Efficiency, with the advisor outperforming the traditional search methods in all aspects. This demonstrates the efficacy and accuracy of RAG based methods over traditional search methods.

Recently, a study by Jungwon Lee et al. [4] aimed to develop and evaluate two types of specialized Large Language Models (LLMs) for construction safety management knowledge retrieval: a Retrieval-Augmented Generation (RAG) model and a fine-tuned LLM. The study sought to compare the performance of these models with a general-purpose LLM, GPT-4 against a custom RAG model and a fine-tuned LLM. The study successfully demonstrates a measurable reduction in hallucinations of the general-purpose LLM by demonstrating an increase in evaluation metrics like BLEU and ROUGE scores. The RAG model was created by integrating GPT-4 with the developed knowledge graph using the LangChain module. Tailored prompts were designed to extract keywords from accident synopses and formulate Cypher queries. The fine-tuned LLM was a Koalpaca-Pyglot-12.8B model, a Korean language understanding model, that was fine-tuned using a question-answering (QA) dataset. The evaluation metrics consisted of metrics such as cosine similarity, Euclidean distance (calculated using embeddings from BERT and OpenAI models), BLEU score, and ROUGE score. The research found that both the RAG model and the fine-tuned LLM significantly outperformed general-purpose GPT-4 in construction safety knowledge retrieval, with improvements of 21.5% and 26%, respectively. The RAG model excelled in generating accurate, verifiable responses using precise terminology from the knowledge graph, while the fine-tuned LLM provided adaptable, multi-perspective answers and better alignment with domain-specific language. Quantitative evaluations showed that although cosine similarity and Euclidean distance were similar across models, the RAG and fine-tuned LLM achieved much higher BLEU and ROUGE scores.

A novel paradigm - RAG4CM was proposed by Chengke Wu et al. [5]. A knowledge graph is created by converting project documents, including plain text, tables, and images, into a hierarchical structure. The content within these hierarchical nodes was then chunked using semantic-based methods on the basis of embeddings generated using the BGE-m3 model. A novel retrieval algorithm is also tested. The algorithm compares the semantic similarity between a user query and the embeddings of all information pieces in the knowledge pool using cosine distance. The algorithm then groups these pieces based on their original document nodes and employs a relevance voting strategy, aiming to return the most relevant nodes. Additionally, a user preference learning algorithm is proposed and tested as a prototype. A local learning mechanism was developed to fine-tune the BGE model based on the user's selections over time. This fine-tuning aims to prioritize information that aligns with individual user preferences in subsequent retrievals. The research found that RAG4CM significantly outperformed existing methods in obtaining construction management information, achieving a high response accuracy (0.898) and a top-3 retrieval accuracy (0.924).

3. Dataset Description

The dataset for our RAG study is built on the Wendel specifications document [10], which details the Office Building Renovation pipeline, including construction processes, codes, and regulations. This document was chosen because it directly addresses the construction safety and management field, making it relevant to our study. The consistent formatting, clear section divisions, and structured headings and subheadings in the document facilitate the segmentation of content and the alignment of Q&A pairs with specific sections. The final dataset is organized in a CSV format containing columns such as "Question Type," "Question," "Answer," and the section of the PDF corresponding to each Q&A pair. The inclusion of varied question types—factual, reasoning, and quantitative—ensures diverse coverage of the domain. Figure 1 represents the entire data preprocessing and ground truth data generation pipeline.

4. Experimental Methodology

In this study, we developed a retrieval-augmented generation (RAG) system for the construction safety and management domain using a ground truth dataset curated from the Wendel specifications document. The dataset, which covers detailed information on the Office

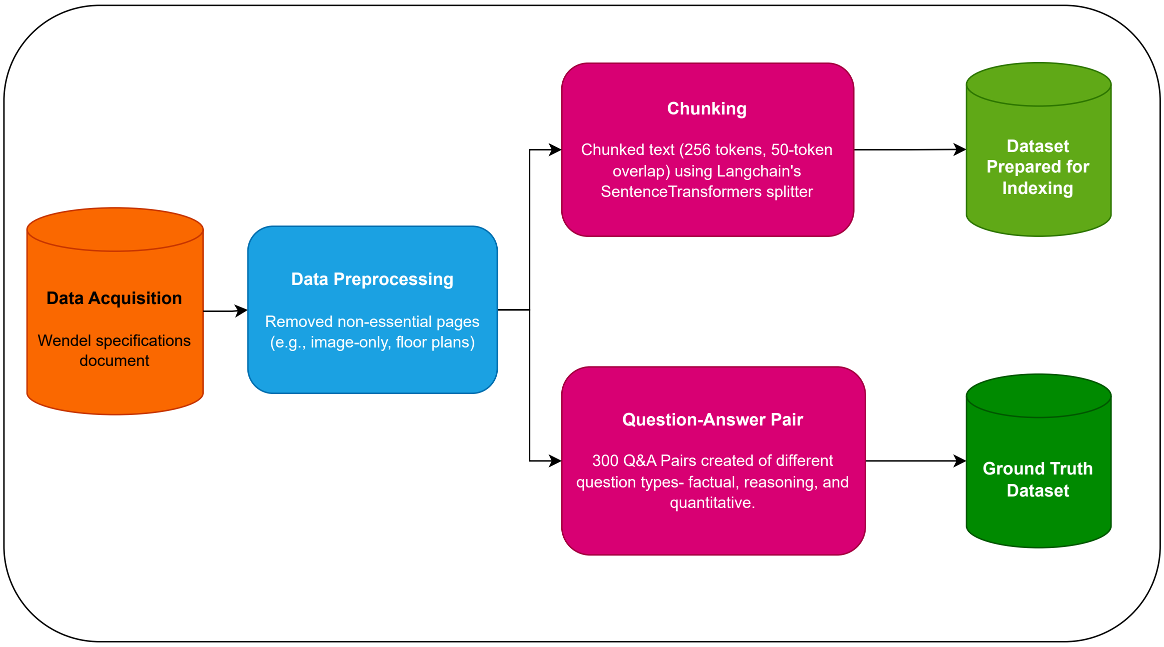

Figure 1: Data Preprocessing and Ground Truth Data Generation Pipeline

Building Renovation pipeline, serves as the basis for generating and evaluating responses to domain-specific queries. Document chunks extracted from the specifications were embedded using state-of-the-art models and then stored and indexed in a Chroma vector database.

Our experimental methodology is organized into several components in the following sections. First, we describe the process of ground truth data generation and data preprocessing. Next, the embedding models used to represent text are detailed, along with their architectural differences and suitability for the task. We then outline the generative models employed which address different trade-offs in context handling and response quality. Finally, the evaluation metrics are introduced as quantitative measures of the generated text quality. An ablation study was conducted by varying the embedding model, generative model, and generation temperature. Figure 2 represents the experimental methodology undertaken during this study.

4.1. Ground Truth Data Generation and Data Preprocessing

For the ground truth data generation, the Wendel specifications document was first divided into three parts, with each team member responsible for generating approximately 100 question and answer pairs per section, resulting in roughly 300 pairs in total. These Q&A pairs were structured into a CSV file with columns for “Question Type,” “Question,” “Answer,” and the corresponding section of the document.

Prior to ground truth generation, data preprocessing was performed on the document. Irrelevant pages, such as those containing images or floor plans, were removed to eliminate noise and ensure that only textual content was retained. The remaining text was then segmented using a chunking strategy based on SentenceTransformersTokenTextSplitter from Langchain. The SentenceTransformers-based splitter preserves semantic boundaries better than naïve character or sentence-based splitters, thereby improving the quality of document chunks used in retrieval. The splitter used a token count limit of 256 tokens per chunk with an overlap of 50 tokens. This chunking approach maintained the contextual flow of the document. In addition, basic text cleaning steps were applied, such as removing extra spaces and correcting encoding issues, to further improve the quality of the text.

The data (the chunks and the ground truth dataset) can be found at Dataset Link.

4.2. Embedding Models

In our study, we employed two text embedding models to encode document chunks into fixed-length vector representations. The first model is the IBM Granite

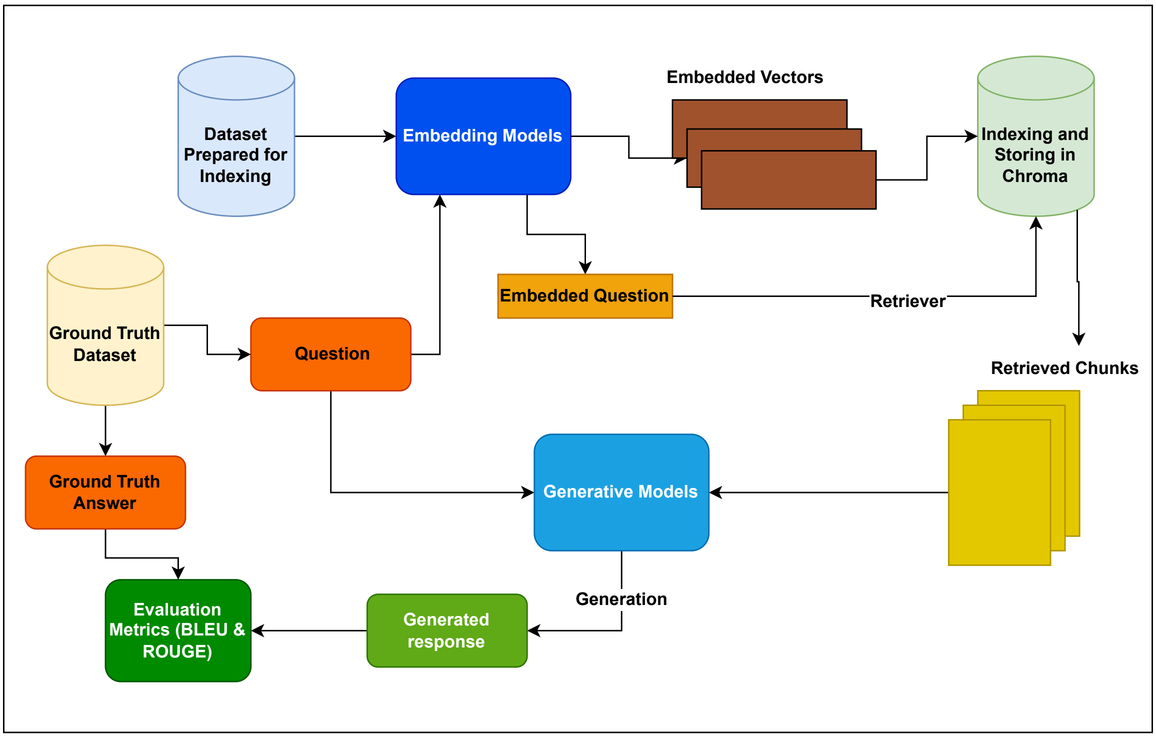

Figure 2: Experimental Methodology

Embedding 278M [6], a text-only dense biencoder. The model produces 384-dimensional embeddings using a BERT-based architecture. Key configurations of the IBM Granite Embedding model include a block count of 6, a context length of 512 tokens, and the use of CLS pooling for generating the final representation.

The second model used in our experiments is nomic-embed-text:latest [7], a large-context-length text encoder. Built on a "nomic-bert" architecture, the model features a deeper configuration with 12 blocks, a context length of up to 2048 tokens, and a 768-dimensional embedding length. Its attention mechanism is configured with 12 heads, and a mean pooling strategy is applied to produce the final embeddings, making it suitable for tasks that require understanding longer passages.

We selected these two embedding models to compare their effectiveness in our retrieval-augmented generation system for the construction safety and management domain. The IBM Granite Embedding 278M offers a proven and efficient solution with its BERT-based architecture. In contrast, nomic-embed-text:latest is designed to handle longer contexts and has demonstrated superior performance on tasks that involve complex, lengthy inputs. Evaluating both models allows us to analyze the trade-offs between model complexity, context handling, and retrieval accuracy.

4.3. Generative Models

In our experiments, we evaluated three different generative models to study the effect of model capacity and context window size on the quality of generated responses. The chosen models are based on the Gemma and DeepSeek families, each with distinct design characteristics and capabilities tailored to tasks such as question answering and summarization.

The first model is "gemma3:1b" [8]. It is a lightweight, multimodal model built on Gemini technology with a 1 billion parameter configuration. It supports a 32K token context window and is designed to handle text tasks. The gemma3:1b model employs a BERT-inspired architecture, featuring 4 attention heads per layer (with 1 key-value head), a block count of 26, and an embedding length of 1152. This model was chosen for its compact design, which makes it suitable for resource-constrained environments. Its ability to process text within a limited context window is particularly useful for generating responses from retrieved contexts that are concise yet informative.

The second model is "gemma3:4b" [8], which is a multimodal model with a 4 billion parameter configuration and an extended 128K token context window. This model is designed for applications that require handling much longer contexts and even has vision components that enable the processing of visual information. For our text-only tasks, the gemma3:4b model utilizes 8 attention heads (with 4 key-value heads) and a block count of 34, with an embedding length of 2560. Its increase in context capacity allows for the inclusion of more detailed retrieved information, which might help with complex construction domain queries where a larger context might capture more relevant details.

The third model in our study is "deepseek-r1:7b" (DeepSeek-R1-Distill-Qwen-7B) [9]. This model exemplifies an approach in which reasoning patterns learned by large models are distilled into a smaller model format. Based on the Qwen-2.5 series architecture, deepseek-r1:7b is fine-tuned using a dataset of 800K samples curated by the DeepSeek team. It features 7 billion parameters and maintains a context window of 131072 tokens. Its architecture has 28 attention heads and an embedding length of 3584.

Collectively, these models were chosen to cover a broad spectrum of design philosophies: from lightweight, context-limited models to heavy, context-extensive architectures. By comparing responses generated by gemma3:1b, gemma3:4b, and deepseek-r1:7b, we aim to understand the trade-offs in model capacity, context handling, and output quality in the construction safety and management domain.

4.4. Evaluation Metrics

In our experiments, we use two well-established metrics to assess the quality of the generated responses: the Bilingual Evaluation Understudy (BLEU) score and the Recall-Oriented Understudy for Gisting Evaluation (ROUGE) score. The BLEU score quantifies the degree of n-gram overlap between the generated text and the reference (ground truth) answer. It is widely used in machine translation and text generation tasks to measure precision aspects, namely, how many of the generated n-grams appear in the reference text. A higher BLEU score indicates that the generated answer is more closely aligned in wording and structure with the ground truth.

Complementing BLEU, the ROUGE score focuses on recall by measuring how many n-grams of the reference text are captured in the generated response. ROUGE also accounts for various levels of granularity, such as ROUGE-N (for n-gram overlap) and ROUGE-L (based on the longest common subsequence).

Using ROUGE in conjunction with BLEU allows us to evaluate both the precision and the completeness of the information in the generated text. These metrics are particularly suitable for our RAG task because they offer a balanced view of both the linguistic accuracy and the completeness of the generated responses.

4.5. Ablation Study Settings

To understand the impact of different model architectures and parameters, we conducted an ablation study across multiple dimensions. The study considered variations in the embedding model, the generative model, and the temperature parameter of the generative model. Two embedding models were evaluated: the nomic-embed-text (137M) and the IBM Granite Embedding Model (278M). The purpose of including these two models is to assess how the quality of text embeddings influences the retrieval effectiveness in our RAG system.

For the generative component, we compared three different models: Gemma3 4b, Gemma3 1b, and DeepSeek-R1-Distill-Qwen-7B. These models vary significantly in size and architecture. By testing these generative models, we aim to determine which configuration offers the best trade-off between coherence, correctness of the response, and resource consumption.

The generation temperature, a parameter that affects the randomness of the output, was varied at three levels: 0.1, 0.5, and 0.8. Lower temperatures (e.g., 0.1) tend to produce more concise and focused responses, whereas higher temperatures (e.g., 0.8) allow for greater diversity.

The settings above were varied to obtain 18 different configurations to conduct the ablation study.

5. Results and Discussion

The ablation study presented evaluates the performance of our RAG system by varying three principal components: the text embedding model, the generative model, and the generation temperature. Quantitative assessments were performed using BLEU and ROUGE metrics. These metrics capture both the precision (n-gram overlap) and recall (coverage of key terminology) of generated answers relative to the ground truth dataset derived from the Wendel specifications document. The results of the ablation study are presented in Table 1.

Across the board, the IBM Granite Embedding Model consistently yielded higher BLEU and ROUGE scores compared to nomic-embed-text. For instance, when coupled with Gemma3 1b at a generation temperature

Sr. No.	Embedding Model	Generative Model	Temperature	BLEU Score	ROUGE Score
1	nomic-embed-text 137M	Gemma3 4b	0.1	0.0885	0.2336
2	nomic-embed-text 137M	Gemma3 1b	0.1	0.0793	0.2733
3	nomic-embed-text 137M	DeepSeek-R1-Distill-Qwen-7B	0.1	0.0571	0.2031
4	nomic-embed-text 137M	Gemma3 4b	0.5	0.1014	0.2528
5	nomic-embed-text 137M	Gemma3 1b	0.5	0.1001	0.2796
6	nomic-embed-text 137M	DeepSeek-R1-Distill-Qwen-7B	0.5	0.0576	0.2117
7	nomic-embed-text 137M	Gemma3 4b	0.8	0.0740	0.2194
8	nomic-embed-text 137M	Gemma3 1b	0.8	0.0763	0.2641
9	nomic-embed-text 137M	DeepSeek-R1-Distill-Qwen-7B	0.8	0.0620	0.2097
10	IBM Granite Embedding Model 278M	Gemma3 4b	0.1	0.1035	0.3000
11	IBM Granite Embedding Model 278M	Gemma3 1b	0.1	0.1138	0.3127
12	IBM Granite Embedding Model 278M	DeepSeek-R1-Distill-Qwen-7B	0.1	0.0715	0.2411
13	IBM Granite Embedding Model 278M	Gemma3 4b	0.5	0.1181	0.3116
14	IBM Granite Embedding Model 278M	Gemma3 1b	0.5	0.1252	0.3178
15	IBM Granite Embedding Model 278M	DeepSeek-R1-Distill-Qwen-7B	0.5	0.0722	0.2422
16	IBM Granite Embedding Model 278M	Gemma3 4b	0.8	0.1053	0.3008
17	IBM Granite Embedding Model 278M	Gemma3 1b	0.8	0.0889	0.2819
18	IBM Granite Embedding Model 278M	DeepSeek-R1-Distill-Qwen-7B	0.8	0.0639	0.2299

Table 1: Ablation Study Results

of 0.5, the IBM model achieved the highest BLEU score (0.1252) and ROUGE score (0.3178). This improvement suggests that the IBM model's BERT-based architecture, which generates 384-dimensional embeddings, better captures the semantic relationships inherent in construction management texts. These findings support similar conclusions in the literature where embedding quality influences retrieval accuracy, for example, in studies that emphasize the importance of strong semantic representation for effective downstream question-answering performance.

The Gemma3 models, regardless of size, performed similarly in several configurations. Interestingly, the compact Gemma3 1b model occasionally outperformed Gemma3 4b, particularly when combined with the IBM embedding at moderate temperatures, suggesting that a lighter model might offer benefits in terms of focused response generation without introducing noise from overly extensive context windows. Conversely, the DeepSeek-R1-Distill-Qwen-7B consistently lagged.

Temperature tuning also emerged as a critical factor. A moderate temperature setting (0.5) generally provided an optimal balance between determinism and diversity in the output. Lower temperatures (0.1) resulted in more deterministic but potentially less comprehensive responses, whereas higher temperatures (0.8) introduced too much variability and led to a reduction in both precision and recall scores.

The experimental outcomes align well with the literature surveyed in the report. Previous studies, such as those by Jungwon Lee et al. [4] and Chengke Wu et al. [5], have demonstrated that carefully tuned RAG systems can reduce hallucinations and increase precision by utilizing embedding representations and appropriately sized generative models.

6. Concluding Remarks and Future Work

While the current approach demonstrates promising performance, several gaps offer opportunities for future research. The IBM Granite model, despite its effectiveness, has a shorter context window compared to the nomic model, potentially limiting its ability to handle very long documents or complex construction codes. Future work could explore hybrid or multi-stage retrieval systems that merge long-context and high-quality short-context embeddings. Additionally, the underperformance of the DeepSeek-R1-Distill-Qwen-7B model indicates the need for further fine-tuning, alternative distillation strategies, or model ensembles to improve reasoning capabilities without compromising output fidelity. Broader validation across different construction management documents and real-world queries is also needed to improve the robustness of the system, and there is potential to integrate multimodal data, such as diagrams and blueprints, to further enrich the interpretability and completeness of the responses.

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